A tutorial for calculating field-specific effect size distributions

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Abstract

Effect sizes are useful for understanding the magnitude of study results and for planning new studies using power analysis. But despite their wide usage, effect sizes are often misinterpreted. This is mostly due to an over-reliance on general effect size benchmarks that were not intended for broad application across a range of different research fields. Inaccurate effect size interpretations can consequently lead to incorrect conclusions regarding the magnitude of study results, as well as incorrect sample size estimates, which can increase the likelihood of false positive results. This article introduces the *ESDist* R package, which is designed to calculate empirically derived effect size benchmarks or a range of reliably detectable empirical effect sizes for a specific research question or field of interest by the calculation of effect size distributions (ESDs). This package can be used on data that can be easily extracted from pre-existing meta-analyses to help researchers more accurately plan new studies or to better understand how an individual study might relate to other studies in their field of interest. *ESDist* includes a set of features that make it easy to use in *a priori* power analysis. Moreover, the package includes a feature for estimating effect size benchmarks that account for publication bias and are weighted by effect sizes variances, which addresses existing limitations of using ESDs for study planning or interpretation.

Keywords: R, Effect size, Effect size distribution, Study design, Power analysis

1. Introduction

Effect sizes are important empirical study outcomes (Lakens, 2013) which, unlike *p*-values, can describe the *magnitude* of an effect, rather than only indicating the *presence* of an effect. This means that effect sizes can be used to make inferences about the practical importance of an observed effect (Cohen, 1988). Unlike an *unstandardised* effect size, which describes an effect size on the same scale as the effect was detected (e.g., weight in kilograms), *standardised* effect sizes (e.g., Cohen's *d*) describe the magnitude of an association or difference between groups in a way that does not refer to the original measurement scale. As such, standardized effect sizes can facilitate comparisons between studies, regardless of differences in study design, population sample, or methods.

Despite the benefits of standardised effect sizes, their magnitudes can be difficult to interpret as they do not refer to their original measurement scale. The use of Cohen's (1988) heuristics for small, medium, and large effect sizes for various standardized effect sizes (e.g., Cohen's *d*, Pearson's *r*) has become ubiquitous in the psychological and biomedical sciences for both interpreting study results and planning new studies via *a priori* power analysis. But this was never Cohen's intention (Funder & Ozer, 2019), as he suggested these heuristics as a last resort for *a priori* power analysis (Cohen, 1988). That is, researchers should only use these heuristics when effect size estimates from previous research are not available. These heuristics have no empirical basis, and therefore do not describe effect sizes as small, medium, or large compared to anything in particular (Funder & Ozer, 2019). The use of imprecise heuristics can lead to issues with understanding the magnitude of individual study results, as well as problems with planning appropriately powered studies (Funder & Ozer, 2019).

The aim of this article is to introduce the ESDist package for the statistical software R (R Core Team, 2023) as an empirical approach for calculating effect size benchmarks for use in *a priori* power analysis or for the interpretation of observed effects. We will begin by discussing the limitations of using Cohen's heuristics to interpret study results or to plan new studies.

Subsequently, we will argue that the use of field-specific Effect Size Distributions (ESDs) could be a useful alternative to the use of Cohen's benchmarks. We then discuss how publication bias can influence the calculation of ESDs, consequently inflating effect size benchmarks. Finally, the *ESDist* package for R will be introduced. We demonstrate how *ESDist* can be used for visualising ESDs using various approaches. We also exhibit how to calculate field-specific effect size benchmarks that can also account for publication bias.

1.1. Issues with general effect size heuristics

The use of Cohen's heuristic benchmarks for the interpretation of results and planning new studies is common across many research fields. A "medium" effect size is supposed to approximate the average effect size in a research field (or for a research question), with "small" and "large" effects equidistant from a medium-sized effect. While Cohen's benchmarks may happen to align with the distribution of effects of some fields, they cannot be accurate for *all* research fields as they vary in terms of methods, measures and populations. Consequently, some researchers have developed field-specific effect size benchmarks from effect size distributions (typically based on the 25th, 50th, and 75th percentiles), which suggest that there is considerable variability between fields when it comes to effect size benchmarks. For example, Gignac and Szodorai (2016) suggested Pearson's *r* values of 0.11, 0.19, and 0.29 to correspond to relatively small, typical, and relatively large effect sizes respectively in the individual differences (i.e., trait psychology) literature. Lovakov and Agadullina (2021) suggested Pearson's *r* values of 0.12, 0.24, and 0.41, and Cohen's *d* values of 0.15, 0.36, and 0.65 to correspond to small, medium, and large effect sizes, respectively, in social psychology literature. An overview of some existing effect size benchmarks is presented in figures 1A and 1B. Figure 1A includes six studies that report effect size benchmarks using Cohen's *d* (Cherubini & MacDonald, 2021; Lovakov & Agadullina, 2021; Nordahl-Hansen et al., 2022; Plonsky & Oswald, 2014; Quintana, 2017; Szucs & Ioannidis, 2017). Figure 1B includes seven studies that report effect size benchmarks using Pearson's *r* (Bosco et al., 2015; Brydges, 2019; Gignac & Szodorai, 2016;

Lovakov & Agadullina, 2021; Paterson et al., 2016; Plonsky & Oswald, 2014; Schäfer & Schwarz, 2019). The corresponding tables can be found in the supplementary materials (Supplementary Tables S1 and S2).

1.2. Implications for using general effect size benchmarks for study interpretation and planning

Given this variability of effect size benchmarks across different fields, the use of Cohen's general effect size heuristics can lead to potential misinterpretations of observed effect sizes, and, in *a priori* power analysis, this can lead to study test and design combinations that are statistically underpowered to reliably answer specific research questions (Lakens, 2022). Statistical power, the probability of correctly rejecting the null hypothesis if there is a true effect, is a function of study design, hypothesized effect size, sample size, and alpha level. Statistical power increases as either the sample size or the hypothesized effect size increases. When designing a study, researchers should decide on a sample size that ensures sufficient statistical power, which is typically 80%, although some researchers suggest 90%-95% power instead (e.g., Lakens, 2022). Inaccurate justification of the effect size parameter in *a priori* power analysis can lead to studies that cannot reliably detect a wide range of plausible effect sizes (Brydges, 2019). For example, if a researcher expects to find a 'medium-sized' effect, they might plan for a one-tailed paired-samples *t*-test using Cohen's suggestion for a medium effect size (d = 0.5) with 80% power, which would require a sample size of N = 27. However, if the 'true' medium effect size (i.e., the 50th percentile effect size) for their specific field is d = 0.4, the study would only have 64.8% power, meaning that there would be a considerable chance of not detecting an effect even though it exists.

There are several ways in which effect sizes can be selected more accurately when performing an *a priori* power analysis. Cohen (1988) originally proposed that the effect size parameter in *a priori* power analysis should be treated as an expected population effect size. However, since the population effect size is almost always unknown, authors have more recently suggested that researchers should treat this parameter as a smallest effect size of interest (SESOI)

instead (Lakens & Evers, 2014). With this approach, the researcher decides on the smallest effect size that is deemed 'worthwhile' and designs a study that is adequately powered to detect this effect size. Consequently, the power level is sufficiently high to detect this smallest effect size of interest, and power only increases for larger effect sizes. A different, yet related, approach is sensitivity power analysis, where the researcher uses a given sample size as a starting point instead of an effect size of interest (Lakens, 2022). This approach is useful when data is already collected, or there is a known limit to the number of participants that can be tested, for example due to resource constraints. In this case, the goal is to gain insight into what *range* of effect sizes can be detected for a range of acceptable power levels. Consequently, the researcher aims to strike a balance between which effect sizes are meaningful to detect or reject and with what level of accuracy the researcher wishes to detect or reject these effects. However, even with these more sensible approaches to power, researchers need to have a good understanding of what effect sizes are interesting for their research question in order to justify the effect size parameter in their power analysis (Lakens, 2022).

1.3. The calculation of field-specific benchmarks using ESDs

Although field-specific benchmarks like those presented in Figures 1A and 1B are likely to outperform Cohen's more general heuristics, the fields for which these heuristics have been developed are still very broad (e.g., "Psychology" or "Medical Science"). Moreover, such benchmarks are not widely available, especially for more specific fields. In many cases, researchers would benefit from having access to an ESD that is specific to their field, sub-field, or even to a particular research question. A field-specific ESD would allow researchers to better understand the magnitude of their observed effect compared to previous research within the field, either by calculating field-specific benchmarks, or by calculating the percentile that corresponds to their observed effect size.

Having access to an ESD might be even more useful when planning new studies. For example, researchers might have specific reasons for setting their SESOI to correspond to a particular percentile in their ESD, such as any of the quartiles that conventionally correspond to

small, medium, or large effect size benchmarks. Alternatively, if a researcher knows their sample size parameter, they can perform a sensitivity power analysis which creates a range of reliably detectable effect sizes. Subsequently, they can use an ESD to calculate the proportion of empirical effect sizes in the ESD that fall within this range for each level of power. Knowing the range of empirical effect sizes that can be reliably detected given a particular study design is useful in several cases. For example, if the SESOI is so high that the percentage of effect sizes that can be detected by the study is close to 0%, it is unlikely that the study will provide any new information to the field (e.g., Lakens et al., 2018).

1.4. Existing limitations of field-specific ESDs

Because field-specific ESDs largely consist of effect sizes reported in published studies, it is likely that they are influenced by publication bias (Button et al., 2013; Lakens, 2022). Publication bias refers to the phenomenon where studies with statistically significant results are more likely to be published than studies without a significant result, which, in turn, leads to an overall inflation of effect sizes (Kvarven et al., 2020; Schäfer & Schwarz, 2019). This is an issue that mainly affects small studies (Dwan et al., 2008). Non-significant effects in small studies are often attributed to low power and are therefore deemed inconclusive or uninformative (Evangelou et al., 2012). The publication of small sample studies with non-significant results is decided against (either by journal editorial teams or the authors not submitting the study for publication in the first place) more frequently than large studies with non-significant results, since non-significant results in studies with a large sample size are more difficult to explain on the basis of low power (Button et al., 2013). Meanwhile, small studies that do find effects are more likely to be published, despite the higher likelihood that they greatly overestimate the true effect size (Rochefort-Maranda, 2021). Aside from publication bias, small studies are more sensitive to other biases, such as vibration effects (i.e., the variability in a study's effect size estimate based on the selection of analysis method), selective data analysis and selective reporting, and reduced quality of other aspects of the study design (Button et al., 2013). Given these

various biases, any estimate of the population effect size that is based on previous literature is likely to be an overestimation in most cases (Lakens, 2022). Therefore, it is essential to address these biases when interpreting individual study results or planning a new study using an ESD.

Another potential issue with ESDs is that effect sizes are not equally informative. The effect size of each study is merely an estimate of the "true" effect size. As such, each effect size also has a standard error, which is a measure of the precision of the effect size estimate. ESDs are not currently calculated with a consideration of effect size precision. This means that effect sizes influence an ESD calculation equally, regardless of their precision. Because more precise estimates are more informative than less precise estimates, it can be desirable to assign more weight to these estimates. While the weighting of effect sizes is commonplace in meta-analysis via inverse variance weighting, this is not a consideration when ESDs are typically calculated.

2. The *ESDist* package for R

Although ESDs are useful for interpreting effect sizes and for planning studies, they are not widely available. One reason for the low availability of ESDs might be the amount of effort required for gathering effect size estimates of individual studies within a specific field of interest. Moreover, even if many effect size estimates are readily available, researchers might find it difficult or time consuming to gather all relevant information from their ESD, such as effect size benchmarks, the range of reliably detectable effect sizes, or the weighted distribution of empirical effect sizes. Furthermore, even if researchers can obtain field-specific ESDs, publication bias in the ESD might render any information gained from the ESD inaccurate.

To this end, the aim of this paper is to introduce the *ESDist* package for the R statistical software suite (R Core Team, 2023). The *ESDist* package was designed to facilitate the calculation and visualisation of field-specific ESDs based on data that can easily be retrieved from meta-analyses, for the purpose of interpreting individual effect sizes and planning new studies. Moreover, by combining the *ESDist* package with the *meta* (Balduzzi et al., 2019) and *metasens* (Schwarzer et

al., 2015) packages, users can justify their SESOI based on an ESD that is adjusted for publication bias. In the following sections we will explore the *ESDist* package by demonstrating its functions accompanied by their respective outputs. Moreover, reasons for why the use of the *ESDist* package may be preferential over other benchmark-based methods, as well as potential limitations and areas of future investigation and development, will be discussed.

2.1. Package overview

The *ESDist* package contains functions for visualising ESDs, as well as a function for calculating effect size benchmarks. The *ESDist* package requires R version 2.10 or higher and can be loaded using the following commands:

```
R>    library(devtools)
      install_github("berntgl/ESDist")
      library(ESDist)
```

These commands first load the *devtools* package (Wickham et al., 2022), which is necessary to download and install packages from GitHub. The *ESDist* package is then downloaded from GitHub and loaded into the user's R environment. An overview of all uses and functionalities of each main function can be found in Figure 2.

2.2. Data files

The package contains two example data files: *ot_dat_raw* and *ot_dat*. The data file *ot_dat_raw* contains 616 effect sizes from 15 meta-analyses on oxytocin intervention studies (Bakermans-Kranenburg & van IJzendoorn, 2013; Chen et al., 2021; Huang et al., 2021; Kang et al., 2022; Keech et al., 2018; Leppanen et al., 2017, 2018; Leslie et al., 2018; Ooi et al., 2017; Peled-Avron et al., 2020; Sabe et al., 2021; Wang et al., 2019; Yang et al., 2021; Zheng et al., 2019; Zhou et al., 2021). Oxytocin is a hormone and neuromodulator that is primarily produced in the hypothalamus, which has been reported to have a role in social and non-social behaviour and cognition. It has been the subject of

comprehensive research interest due to its role in social behaviour, cognitive regulation, and metabolic homeostasis (Jurek & Neumann, 2018; Quintana & Guastella, 2020).

Some studies have been included multiple times in the dataset, as they reported multiple effect sizes. Thus, the package contains a second, processed data file, *ot_dat*, where duplicates have been removed, and all effect sizes have been converted to the same metric (Hedges' *g*). In cases where a study was included twice or more in the original dataset, only the effect size with the smallest standard error was considered. In the unlikely case that the standard error was the same between two effect sizes of the same study, only the smallest effect size (i.e., the one closest to 0, regardless of the direction) was considered. As a result, the updated dataset contains 183 effect sizes. Both datasets contain a column with effect sizes, *yi*. All effect sizes in this column are reported as Hedges' *g* in the *ot_dat* dataframe, or as the original effect size in the *ot_dat_raw* dataframe. The dataset also contains a column, *sei*, that contains the standard error of the effect size. The dataset includes information on each study, such as sample size (*n_total*), study year (*study_year*), study design (*design*), oxytocin dosage (*doses*), the clinical diagnosis of the subsample (*group*), and details about the meta-analysis from which the study originated (*meta_analysis*, *meta_analysis_doi*, *meta_analysis_year*).

Users can get an overview of all the included columns and the data they contain by using the following command:

```
R>    head(ESDist::ot_dat)
```

2.3. Analysis script and data filtering script

To aid users in using the *ESDist* package, an example analysis script (*analysis_script.R*) is included in the package, which contains examples of all functions and their features. This script, as well as a script (*data_filtering.R*) for processing the raw data (*ot_dat_raw*) to obtain the filtered dataset (*ot_dat*) are included in the supplementary materials can be accessed through https://github.com/berntgl/*ESDist*.

2.4. Publication bias adjustment

The *ESDist* package contains two functions, *esd_plot_pba()* and *esd_table_pba()*, that allow users to adjust their ESD for publication bias. These functions make use of limit meta-analysis (Rücker et al., 2011; Schwarzer et al., 2015). Limit meta-analysis uses a random-effects model that includes a parameter that accounts for small-study effects for the potential adjustment of individual effect sizes, which makes it a suitable adjustment method for ESDs.

2.5. Basic plotting function

The most basic plotting function in the *ESDist* package, *esd_plot()*, takes three basic arguments to create an ESD plot: a data frame (*df*), the column within that data frame that contains all the effect sizes (*es*), and a string that describes the type of effect size (*es_type*). The *esd_plot()* function takes these arguments and outputs a *ggplot* object. The code below demonstrates the function using the *ot_dat* data frame to generate the plot in Figure 3. The column in the *ot_dat* dataset that contains all effect sizes is entitled *yi*, and the effect size measure is Hedges' *g*.

```
R>    esd_plot(
            df = ot_dat,
            es = yi,
            es_type = "Hedges' g")
```

In order to plot the weighted distribution, which is calculated using inverse variance weighting, the user needs to add a *se* argument, which corresponds to the column in the dataset which contains all the standard errors and set the *weighted* argument to *TRUE* in order to generate the plot in Figure 4.

```
R>    esd_plot(
            df = ot_dat,
            es = yi,
```

```
            se = sei,
            weighted = TRUE,
            es_type = "Hedges' g")
```

Either the weighted or the regular ESD can be used with additional functionality. By setting the *method* argument to *"quads"*, the user can plot the small, medium, and large effect size benchmarks based on the 25th, 50th, and 75th percentiles, respectively (or on the 16.65th, 50th, and 83.35th percentiles if set to *"thirds"*). Alternatively, if the user wants to plot the range of empirical effect sizes that falls above or below an effect size of interest, the user can provide the *sesoi* argument with a numerical value. For example, if the user finds that the smallest effect size they can reliably detect with a given level of power is *g* = 0.3, the user may set the *sesoi* argument to 0.3. The returned plot will then visualise what percentage of the ESD is smaller than the 0.3, and which percentage is equal to or greater than the given value. This can be done by using the following code:

```
R>    esd_plot(
            df = ot_dat,
            es = yi,
            sesoi = 0.3,
            es_type = "Hedges' g")
```

2.6. Other plotting functions

The other plotting functions, *esd_plot_group()* and *esd_plot_pba()*, work similarly to the basic *esd_plot()* function by taking the *df*, *es*, and *es_type* arguments. The *esd_plot_group()* function allows users to plot ESDs for each group, and therefore requires an extra argument, *grouping_var*, which corresponds to the column in the dataset on which the data should be grouped. By default, it

will only plot groups with at least 20 effect sizes. An example is showed in Figure 5, which is generated by the following code:

```
R>    esd_plot_group(
            df = ot_dat,
            es = yi,
            grouping_var = group,
            es_type = "Hedges' g")
```

The *esd_plot_pba()* function allows users to plot an "iceberg" plot, which shows the unadjusted ESD with the ESD adjusted for publication bias in the negative direction. For this function, users need to provide an additional *se* argument (see above). The publication bias adjustment is done using limit meta-analysis (Rücker et al., 2011), which assesses the relationship between effect size and study size and corrects for this potential correlation. An example of an iceberg plot is shown in Figure 6, which is generated using the following code:

```
R>    esd_plot_pba(
            df = ot_dat,
            es = yi,
            se = sei,
            es_type = "Hedges' g")
```

2.7. Calculating effect size benchmarks in tables

The *esd_table()* function is the primary table function in the *ESDist* package and can be used to calculate field-specific effect size benchmarks. In its most basic form, it takes two arguments: a data frame (*df*) and the column within that dataset which contains effect sizes (*es*). Like the plotting functions, the *esd_table()* function also takes a method argument which can be set to either *"quads"* or *"thirds"*. However, unlike the plotting functions, the method argument does not have to be

specified by the user, as it defaults to *"quads"*. The code below sets the *df* argument equal to the *ot_dat* data frame and the *es* argument equal to the column within that data frame which contains effect sizes, *yi*. The output of this function is presented in Table 1.

```
R>    esd_table(
            df = ot_dat,
            es = yi)
```

Like the plotting functions, the table functions can base their benchmark estimates on the weighted distribution by providing the *se* argument with the name of the column containing the standard errors and by setting the *weighted* argument to *TRUE*. The resulting table is reported in Table 2.

Another feature of the *esd_table()* function is that it can calculate benchmarks per group. The code below specifies that benchmarks should be calculated for each diagnosis group in the *ot_dat* dataset with at least three effect sizes (the minimum number of effect sizes necessary to calculate three benchmarks), by setting *grouping_var* to *group*. The resulting table is reported in Table 3. Note that the last row of the table summarises the benchmarks for all effect sizes included in the dataset.

```
R>    esd_table(
            df = ot_dat,
            es = yi,
            grouping_var = group)
```

2.8. Calculating publication bias-adjusted effect size benchmarks

The second table function is the *esd_table_pba()* function, which calculates the raw benchmarks, as well as the publication bias-adjusted benchmarks (using limit meta-analysis; Rücker et al., 2011). As with the *esd_table()* function, the *esd_table_pba()* function takes the standard *df* and *es* arguments,

but takes an additional *se* argument, which corresponds to the column of the dataset that contains the standard errors. The code below generates Table 4.

```
R>    esd_table_pba(
            df = ot_dat,
            es = yi,
            se = sei)
```

The *esd_table_pba()* function allows users to calculate the publication bias adjusted benchmarks per group by providing a *grouping_var* argument to the function, like in the *esd_table()* function. It also allows users to retrieve benchmarks based on the weighted unadjusted and adjusted ESDs by adding *weighted = TRUE*.

3. Summary

The *ESDist* package for R is a tool that facilitates the straightforward calculation of field-specific effect size benchmarks and ESD visualisations. Importantly, the package allows users to create ESDs using data that can easily be obtained from meta-analyses, as well as facilitating the adjustment of the effect size distribution for publication bias and effect size precision.

There are two primary use cases for the *ESDist* package. The first is to help plan new studies based on information gained from a field-specific ESD. The *ESDist* package allows researchers to calculate field-specific effect size benchmarks based on a set of percentiles (e.g., the 25th, 50th, and 75th percentiles) that correspond to 'small', 'medium', and 'large' effect sizes, respectively. These benchmarks can be used in place of more general effect size benchmarks or heuristics (such as those proposed by Cohen, 1988) to calculate the sample size required to detect a given effect size with a specified level of certainty. Alternatively, researchers can use the *ESDist* package to calculate the range of effect sizes that can be reliably detected with a given study test and design combination.

This is particularly useful for situations in which there is a known limit to the number of participants that can be tested in a study, as it allows researchers to gauge whether the planned study could make a valuable contribution to its field.

The second use case for the *ESDist* package is for interpreting new study results. For example, researchers can use the *ESDist* package to calculate effect size benchmarks for their specific field of interest to compare new study results against. Alternatively, researchers can avoid using descriptive effect size benchmark labels altogether and report the percentile of the ESD their study result corresponds to. In other words, instead of making statements such as "we found an effect that was small-to-medium in size", researchers can state the specific percentile of the ESD which corresponds to the effect size (e.g., "we found an effect that was larger than 34% of effect sizes reported in the field"). Although this is a more precise approach, it should be noted that situating an individual study within an ESD is by no means a complete interpretation of a study result in and of itself (Panzarella et al., 2021; Pogrow, 2019). Researchers should therefore aim to explain their study results within the context of their study design, as well as describe the position of their study within their respective ESD by comparing their study to other studies within the field.

Publication bias is a pressing issue in the psychological sciences, in which some studies are more likely to be published than others (Button et al., 2013; Lakens, 2022). Consequently, effect size estimates drawn from empirical studies may be inflated. The *ESDist* package allows users to account for publication bias by incorporating limit meta-analysis (Rücker et al., 2011; Schwarzer et al., 2015). The *ESDist* package can be used to visualise the adjusted ESD against the original ESD via an iceberg plot, or to calculate adjusted effect size benchmarks. Using a publication bias-adjusted ESD will give the user a more accurate representation of how effect sizes are distributed and will subsequently lead to more accurate estimations of sample size.

The *ESDist* package can be a useful tool for justifying the parameters of a power analysis or for interpreting new study results. Nevertheless, there are some limitations that are worth mentioning. The package was designed for use with data that can easily be obtained from meta-

analyses, but there are potential drawbacks to this method. This approach assumes that meta-analyses are available for the field of interest, that these meta-analyses are of sufficient size to create an informative ESD, and that these meta-analyses have reported error-free data. It is recommended that users double-check any unrealistically large values when extracting data from existing meta-analyses.

While ESDs with larger samples are desirable, researchers need to be aware that adding studies with too much heterogeneity (e.g., studies using different methods, populations, etc.) can reduce the specificity of the ESD for a particular study. Heterogeneity, in this case, refers to the variety in results obtained from studies that are conceptual or close replications, which makes it difficult to obtain relevant effect size benchmarks or empirical effect size ranges for any particular study (Linden & Hönekopp, 2021). Given that research methods, populations, and measurements might differ substantially between (or even within) scientific fields, there is a good chance that including too many effect sizes could lead to a very heterogeneous pool of empirical effect sizes (Bryan et al., 2021; Higgins & Thompson, 2002; Linden & Hönekopp, 2021). Heterogeneity can be addressed by using the grouping feature included in several functions within the package. It is recommended that users take note of reported heterogeneity of the meta-analyses they retrieved their data from or perform their own test of heterogeneity (Linden & Hönekopp, 2021).

Altogether, the overall goal of this tutorial is to instruct readers on how to generate field-specific effect size distributions for the calculation and visualisation of effect size benchmarks or ranges of detectable empirical effect sizes to aid researchers in planning new studies using the *ESDist* package. The package can be used with data that can be extracted from pre-existing meta-analyses and can help researchers plan informative new studies *a priori* and gain information about how an individual study might relate to other studies in their field of interest by using conventional effect size benchmark percentiles for small, medium, or large effects (i.e., the 25th, 50th, and 75th percentiles), or by using more precise percentage values.

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Competing interests

The authors declare no potential competing interests.

Data availability statement

The data can be accessed as a part of the R package code, which is located at

https://github.com/berntgl/*ESDist.*

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Table 1

The output of the esd_table() function for all effect sizes in the ot_dat dataset

	25 %	50 %	75 %	Number of effects
Raw effect size	0.06	0.23	0.50	182

Note. These are the benchmarks for small, medium, and large effect sizes based on the *ot_dat* dataset. The benchmarks are based on the 25th, 50th, and 75th percentiles, which corresponds to the 'quads' method.

Table 2

The output of the esd_table() function for all weighted effect sizes in the ot_dat dataset

	25 %	50 %	75 %	Number of effects
Raw effect size	0.05	0.20	0.43	182

Note. These are the benchmarks for small, medium, and large effect sizes based on the weighted effect sizes in the *ot_dat* dataset. The benchmarks are based on the 25th, 50th, and 75th percentiles, which corresponds to the 'quads' method.

Table 3

The output of the esd_table() function for all effect sizes in the ot_dat dataframe per group

Group	25%	50%	75%	Number of effects
AN	0.02	0.05	0.06	6
Autism	0.11	0.31	0.55	32
BPD	0.16	1.22	2.29	5
PTSD	0.2	0.32	0.38	6
SCZ	0.05	0.13	0.24	20
Anxiety	0.19	0.31	0.41	4
Depression	0.21	0.45	0.91	6
Neurotypical	0.05	0.25	0.66	89
All	0.06	0.23	0.5	182

Note. These are the benchmarks for small, medium, and large effect sizes for each group in the *ot_dat* dataset with at least 4 effect sizes. The benchmarks are based on the 25th, 50th, and 75th percentiles, which corresponds to the 'quads' method. AN = Anorexia nervosa; BPD = Bipolar disorder; OCD = Obsessive compulsive disorder; PTSD = Post-traumatic stress disorder; SCZ = Schizophrenia.

Table 4

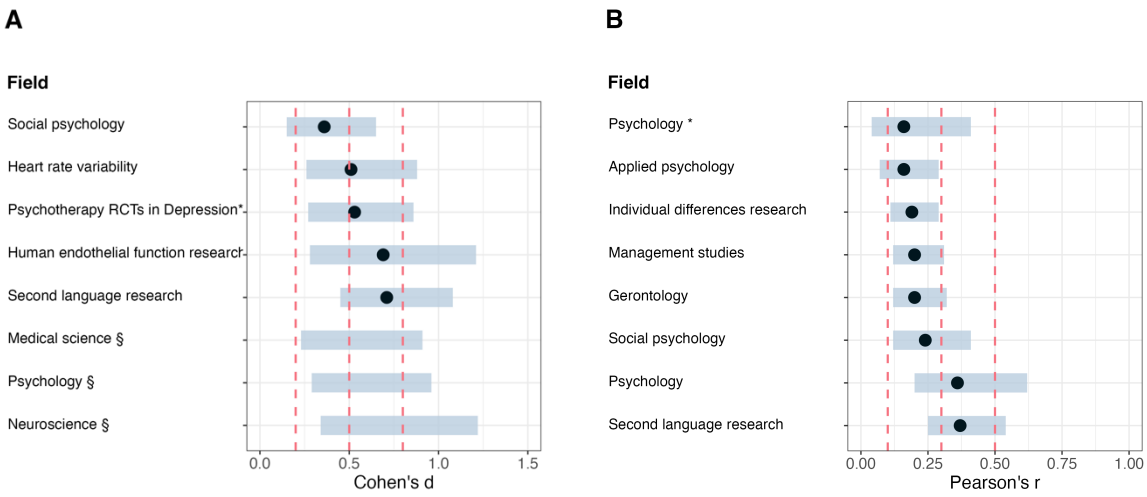
The output of the esd_table_pba() function for all effect sizes in the ot_dat dataset

	25 %	50 %	75 %	Number of effects
Raw effect size	0.06	0.24	0.50	182
Adjusted effect size	0.08	0.20	0.39	182

Note. These are the benchmarks for small, medium, and large effect sizes for each group in the ot_dat dataset with at least 4 effect sizes, as well as the benchmarks adjusted for publication bias. The benchmarks are based on the 25th, 50th, and 75th percentiles, which corresponds to the 'quads' method.

Figure 1

Effect size benchmarks for several fields in the biomedical sciences compared to Cohen's suggested benchmarks

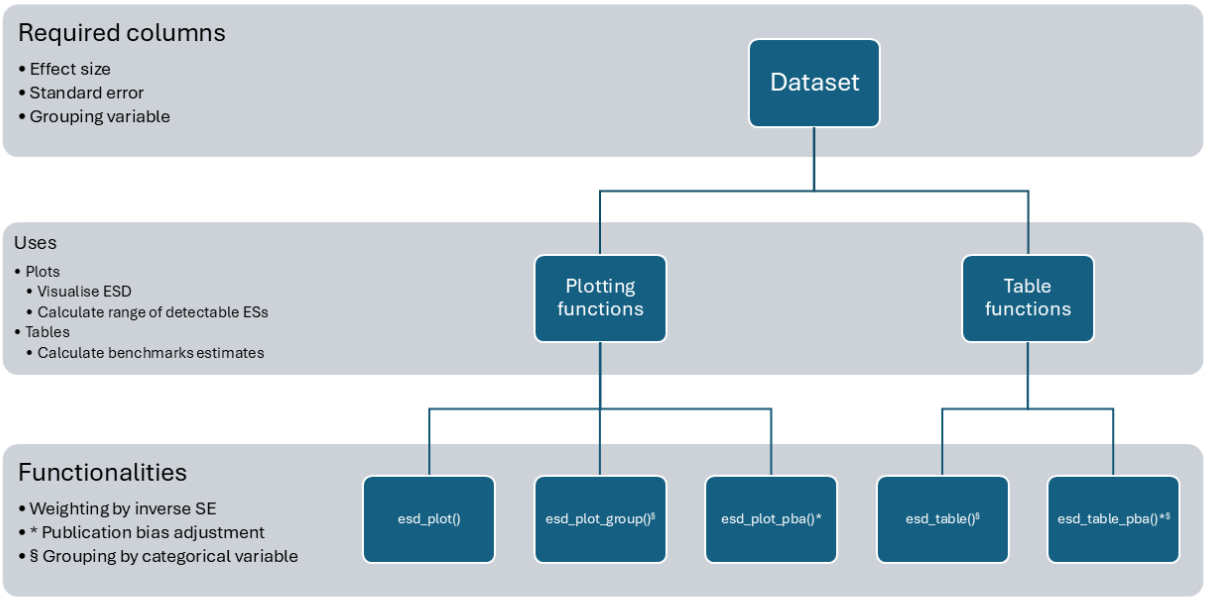


Note. **A)** Effect size benchmarks for several fields in Cohen's *d*. Each bar represents an interquartile range (i.e., all effect sizes between the 25th to 75th percentiles). Each dot represents the median, if reported. The red dashed lines represent Cohen's (1988) heuristics for small, medium, and large effect size, which are *d* = 0.2, 0.5, and 0.8, respectively.

B) Effect size benchmarks for several fields in Pearson's *r*. Each bar represents an interquartile range (i.e., all effect sizes between the 25th to 75th percentiles), except the bars for the field of psychology (taken from Schäfer and Schwarz, 2019), where the bars represent the range from the 16.65th to the 83.35th percentiles. Each dot represents the median. The red dashed lines represent Cohen's (1988) heuristics for small, medium, and large effect size, which are *r* = 0.1, 0.3, and 0.5, Studies with an asterisk (*) have been adjusted for publication bias. Studies with a section sign (§) did not report a median value.

Figure 2

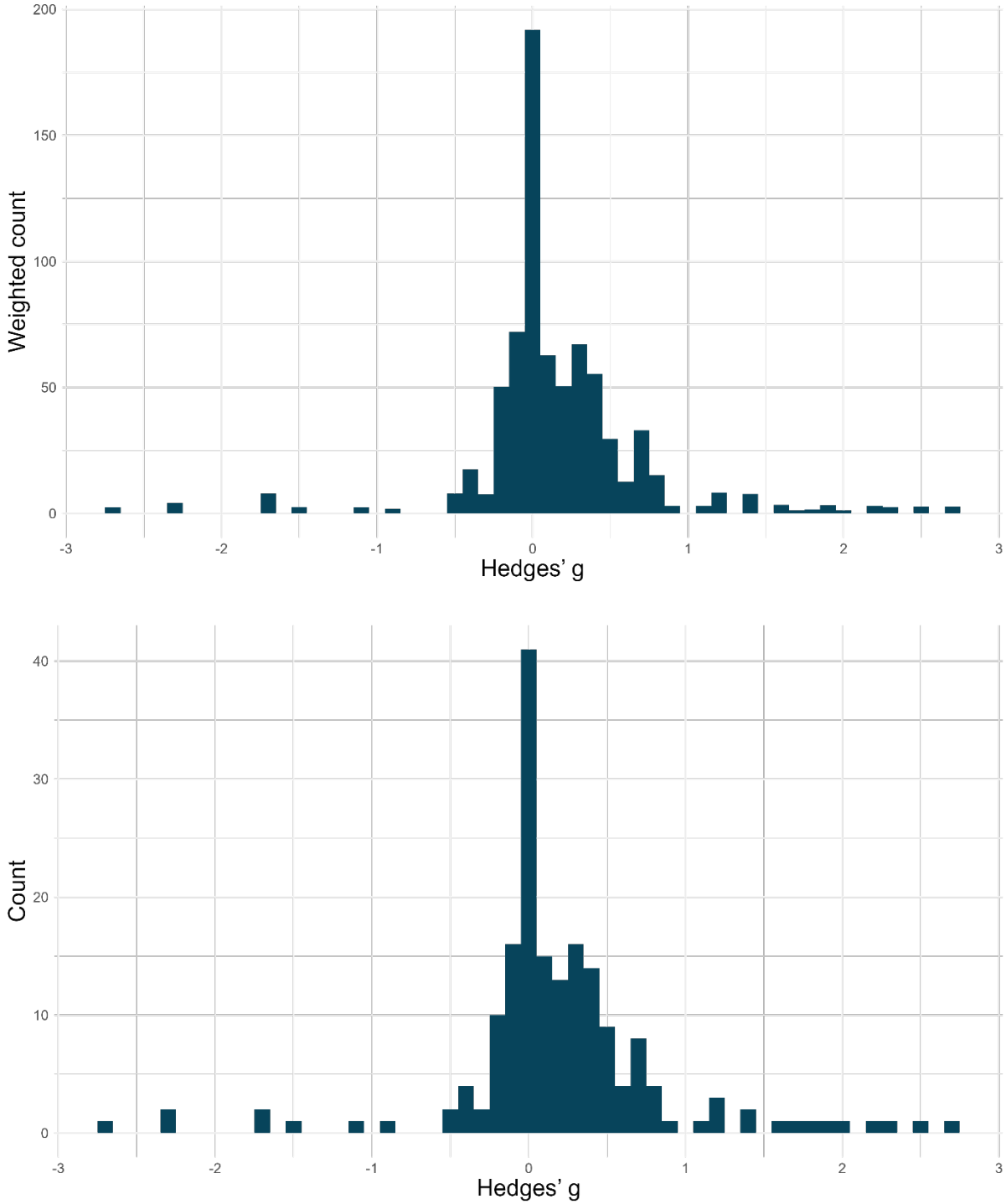
An overview of all main functions in the ESDist package.



Note. An overview of the *ESDist* package for R. The dataset that is to be used with the *ESDist* package should contain at least columns for effect size, standard error, and a grouping variable to make use of all functions in the package. The functions in the package can be divided into tow categories: plotting functions and table functions. The plotting functions are primarily useful for visualising ESDs, as well as for calculating the range of detectable empirical effect sizes, whereas table functions are primarily useful for calculating estimates for effect size benchmarks. Finally, all functions allow for inverse weighting by standard error to make more precise estimates in the dataset more informative. Functions with an asterisk (*) also allow for publication bias adjustment, and functions with a section sign (§) allow for grouping by a categorical variable.

Figure 3

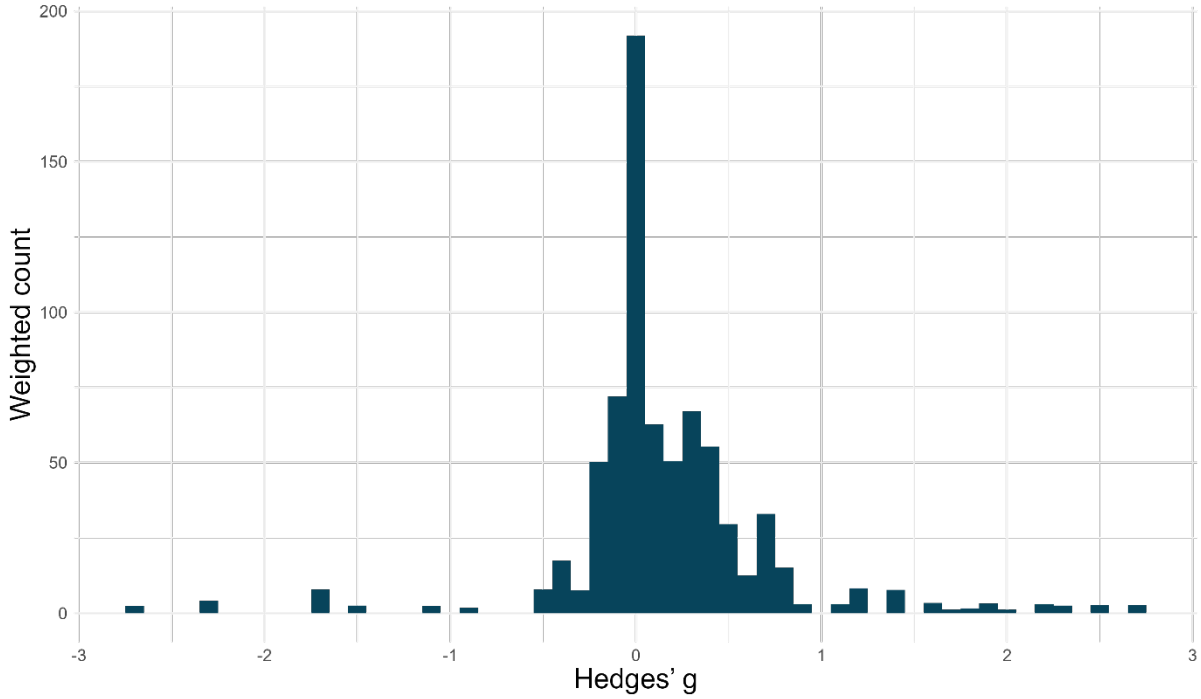
Output of the esd_plot() function showing the distribution of effect sizes from the ot_dat dataframe



Note. The outcome of the *esd_plot()* function. This histogram shows the distribution of effect sizes for the *ot_dat* dataset. Each bin represents the number of effect sizes within an interval of *g* = 0.1.

Figure 4

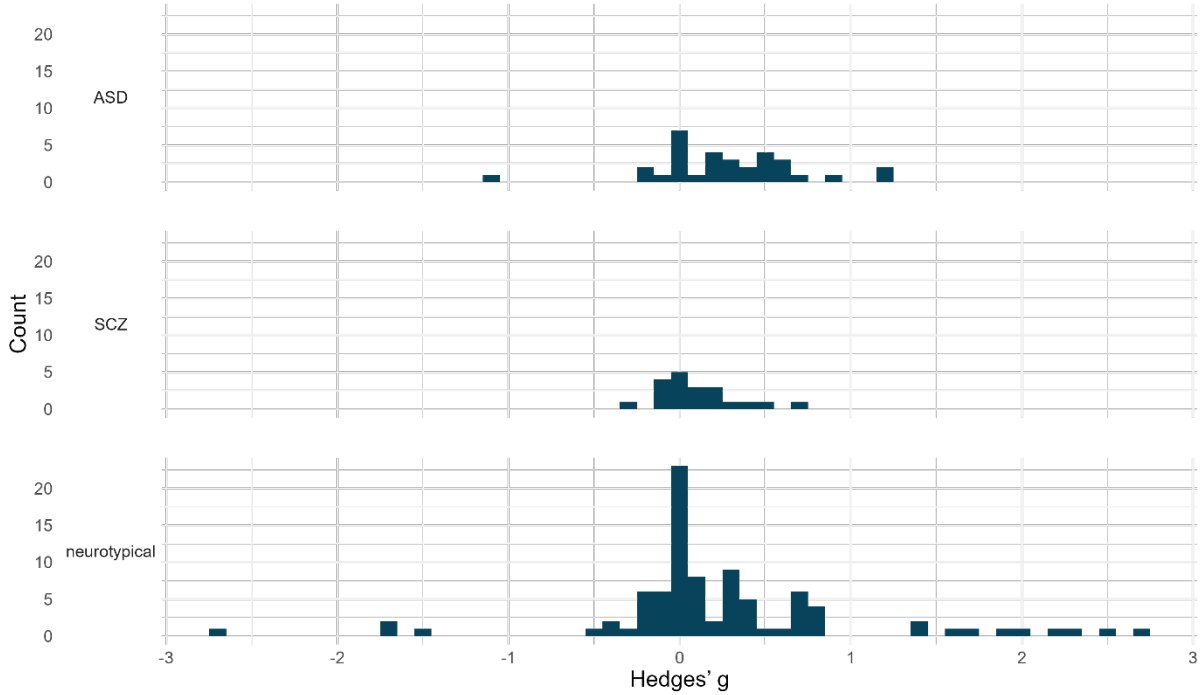
Output of the esd_plot() function showing the weighted distribution of effect sizes for effect sizes in the ot_dat dataframe



Note. The outcome of the *esd_plot()* function. This histogram shows the weighted distribution of effect sizes for the *ot_dat* dataset. Each bin represents the number of effect sizes within an interval of g = 0.1.

Figure 5

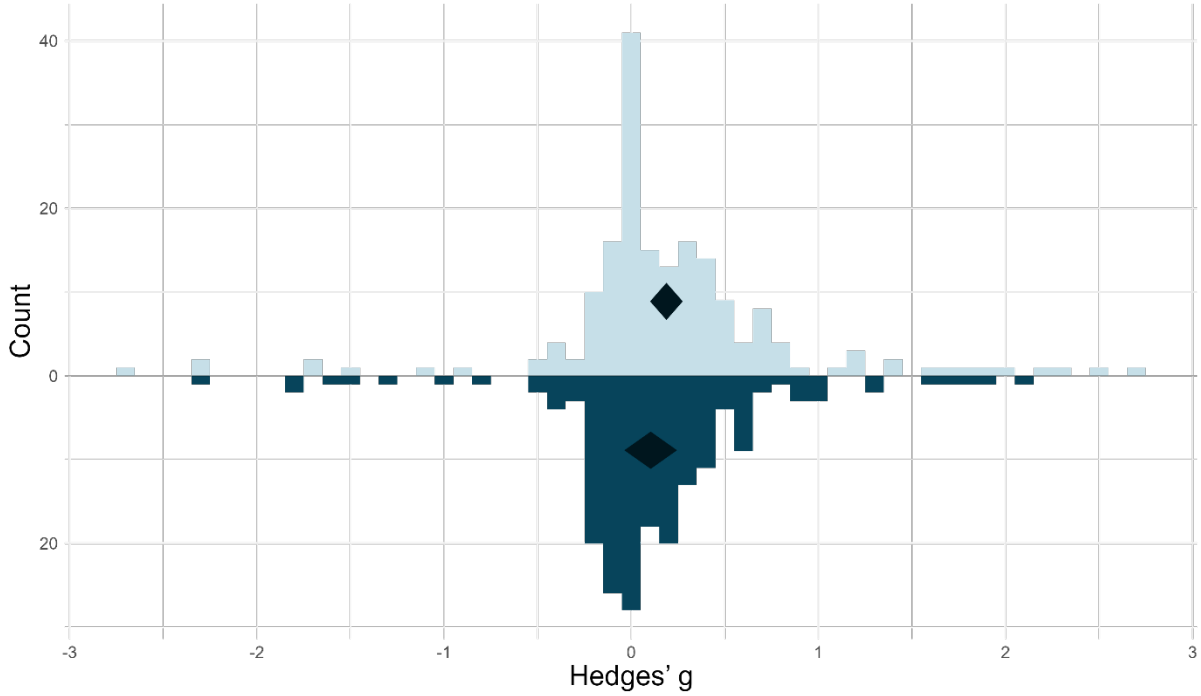
Output of the esd_plot_group() function showing ESDs per group



Note. The outcome of the *esd_plot_group()* function. This figure contains plots for the distribution of effect sizes for each group in the *ot_dat* dataset which contains at least 20 effect sizes. In the case of the *ot_dat* dataset, these are only the autism spectrum disorder, neurotypical and schizophrenia groups. Each bin represents the number of effect sizes within an interval of *g* = 0.1. ASD = Autism spectrum disorder, SCZ = Schizophrenia.

Figure 6

Output of the esd_plot_pba() function showing an iceberg plot



Note. The outcome of the *esd_plot_pba()* function. This iceberg plot shows the distribution of unadjusted effect sizes (top distribution in light blue) as well as the distribution of effect sizes adjusted for publication bias according to the limit meta-analysis (bottom distribution in dark blue). The summary effect sizes with the 95% CI are shown in the black diamonds in each distribution. Each bin represents the number of effect sizes within an interval of g = 0.1. CI = Confidence interval.